Heaven's Light Is Our Guide

Rajshahi University Of Engineering & Technology

Department Of Electrical & Computer Engineering

Lab Report-02

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Experiment No-1: Simplification of $F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$

Problem Definition

Simplification of the four-variable Boolean expression $F(A,B,C,D) = AB + \overline{B}C + AC + \overline{A}B\overline{D}$ using Boolean algebra and Karnaugh map (K-map) techniques, followed by schematic implementation and functional verification of both original and simplified logic circuits using logic simulation in Logisim.

Objective

This experiment focuses on simplifying a four-variable sum-of-products (SOP) Boolean expression down to its minimal SOP form through algebraic theorems, then checking that minimization against a 4-variable Karnaugh map. The reduced and original circuits are also built and tested in Logisim to confirm they behave identically and that the gate count actually drops.

Algebraic Simplification

Boolean Expression Simplification Steps

- Initial Expression: F(A, B, C, D) = $AB + \overline{B}C + AC + \overline{A}B\overline{D}$
- Step 1: Application of Consensus Theorem: The term AC is the consensus term between $AB \text{ and } \overline{B}C\left(AB + \overline{B}C + AC = AB + \overline{B}C\right)$.Eliminating the redundant term AC yields:

$$F = AB + \overline{B}C + \overline{A}B\overline{D}$$

- Step 2: Rearranging and Factoring: Grouping the terms containing B:

$$F = \left(AB + \overline{A}B\overline{D}\right) + \overline{B}C$$

$$F = B\left(A + \overline{A\overline{D}}\right) + \overline{B}C$$

- Step 3: Application of Redundancy Law $\left(X + \overline{X}Y = X +\right)$

$$A + \overline{A\overline{D}} = A + \overline{D}$$

- Step 4: Distributing Back:

$$F = B\left(A + \overline{D}\right) + \overline{B}C$$

$$F = AB + B\overline{D} + \overline{B}C$$

Truth Table

Minterm	A	B	C	D	F
m_0	0	0	0	0	0
m_1	0	0	0	1	0
m_2	0	0	1	0	1
m_3	0	0	1	1	1
m_4	0	1	0	0	1
m_5	0	1	0	1	0
m_6	0	1	1	0	1
m_7	0	1	1	1	0
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	1
m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	1
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	1

Table 1: Truth table for $F = AB + B\overline{D} + \overline{B}C$

Karnaugh Map

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	0	0	1
11	1	1	1	1
10	0	0	1	1

Table 2: K-map for $F(A,B,C,D) = AB + \overline{B}C + AC + \overline{A}B\overline{D}$

Grouping Analysis:

- Quad 1 (AB): Entire row $AB = 11$ (cells $m_{12}, m_{13}, m_{15}, m_{14}$), eliminating C and D, yielding AB.
- Quad 2 ($\overline{B}C$): Columns $CD = 11, 10$ across rows $AB = 00, 10$ (cells m_2, m_3, m_{10}, m_{11}), eliminating A and D, yielding $\overline{B}C$.
- Quad 3 ($B\overline{D}$): Columns $CD = 00, 10$ across rows $AB = 01, 11$ (cells m_4, m_6, m_{12}, m_{14}), eliminating A and C, yielding $B\overline{D}$.

All 10 minterms are fully covered by these three essential prime implicant quads. Therefore, the minimal sum-of-products expression is: $F = AB + B\overline{D} + \overline{B}C$

Circuit Diagram

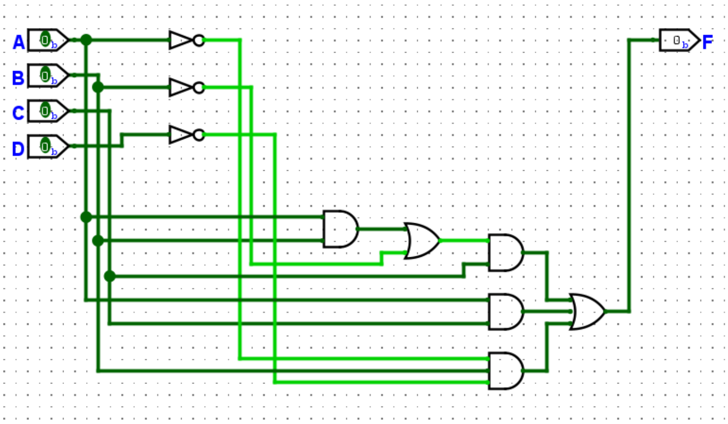


Figure 1:Simplified circuit in Logisim

Discussion

The unsimplified circuit requires three 2-input AND gates, one 3-input AND gate, three inverters (NOT gates), and a 4-input OR gate. The simplified circuit implements $F = AB + B\overline{D} + \overline{B}C$ using only three 2-input AND gates, two inverters (for $\overline{B} and \overline{D}$), and one 3-input OR gate. Both circuits were simulated in Logisim across all 16 input vectors ($0000_2 to 1111_2$). The output states matched for every combination, confirming complete functional equivalence while reducing gate count and propagation delay.

Conclusion

The Boolean expression $F = AB + \overline{B}C + AC + \overline{A}B\overline{D}$ was successfully simplified to its minimal SOP form $F = AB + B\overline{D} + \overline{B}C$. The result was confirmed using a 4-variable Karnaugh map and validated through Logisim logic simulation.

Experiment No-2: Simplification of $F = \overline{(\overline{A} + B)(C + \overline{D})} \cdot \left(A\overline{B} + \overline{\overline{C}D}\right)$

Problem Definition:

Simplification of the nested Boolean expression using Boolean algebra and Karnaugh map techniques, followed by circuit implementation and functional simulation in Logisim.

Objective:

To reduce a complex four-variable Boolean expression containing nested complements into its minimal form using De Morgan's laws and algebraic identities, verify the reduction with a Karnaugh map, and confirm functional equivalence in Logisim.

Algebraic Simplification Steps

1. Applying De Morgan's Law to the First Factor:

$$\overline{(\overline{A} + B)(C + \overline{D})} = \overline{(\overline{A} + B)} + \overline{(C + \overline{D})}$$

$$= \left(\overline{\overline{A}} \cdot \overline{B}\right) + \left(\overline{C} \cdot \overline{\overline{D}}\right) = A\overline{B} + \overline{C}D$$

2. Applying De Morgan's Law to the Second Factor:

   The inner term $\overline{\overline{C}D}$ *expands to* $C + \overline{D}$.

   Therefore, the second factor becomes $\left(A\overline{B} + \overline{C} + \overline{D}\right)$.

3. Combining and Multiplying the Factors:

$$F = \left(A\overline{B} + \overline{C}D\right)\left(A\overline{B} + \overline{C} + \overline{D}\right)$$

$Let X = A\overline{B}$ *and* $Y = \overline{C}D$. *Since* $\overline{Y} = \overline{\overline{C}D} = C + \overline{D}$

the expression fits the form $(X + Y)\left(X + \overline{Y}\right)$

Applying the distributive and complement laws:

$$(X + Y)\left(X + \overline{Y}\right) = X + Y\overline{Y} = X + 0 = X$$

Since $X = A\overline{B}$, *the final simplified expression is*:

$$F = \overline{AB}$$

Truth Table

Minterm	A	B	C	D	F
m_0	0	0	0	0	0
m_1	0	0	0	1	0
m_2	0	0	1	0	0
m_3	0	0	1	1	0
m_4	0	1	0	0	0
m_5	0	1	0	1	0
m_6	0	1	1	0	0
m_7	0	1	1	1	0
m_8	1	0	0	0	1
m_9	1	0	0	1	1
m_{10}	1	0	1	0	1
m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	0
m_{13}	1	1	0	1	0
m_{14}	1	1	1	0	0
m_{15}	1	1	1	1	0

Table 3: Truth table for $F = \overline{AB}$

Karnaugh Map

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

Table 4: K-map for $F(A,B,C,D) = \overline{(\overline{A} + B)(C + \overline{D})} \cdot (\mathrm{A}\overline{B} + \overline{\mathrm{C}}D)$

Grouping Analysis:

- Quad 1 ($\overline{AB}$): All four cells in row $AB = 10$ (m_8, m_9, m_{11}, m_{10}) contain 1s. Grouping this entire row spans all columns, eliminating both variables C and D.

The single quad covers all 1-cells on the map, giving:

$$F = \overline{AB}$$

Circuit Diagram

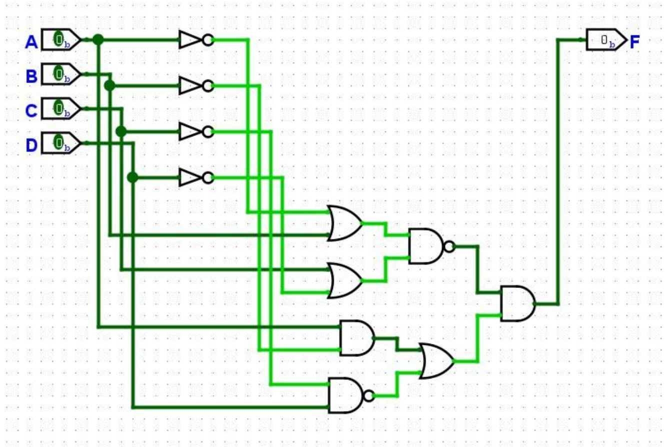


Figure 2:Simplified circuit realizing F = ¯{AB} simulated in Logisim

Discussion

The original circuit requires four OR gates, three AND gates, and five NOT inverters arranged in multiple cascaded logic levels. The simplified circuit implements $F = \overline{AB}$ using only a single 2-input AND gate and one inverter on input B. Both circuits were simulated in Logisim for all 16 input combinations. The output was observed to be logic 1 strictly when $A = 1$ and $B = 0$, completely independent of C and D.

Conclusion

The complex expression $F = \overline{(\overline{A} + B)(C + \overline{D})} \cdot \left(A\overline{B} + \overline{\overline{C}D}\right)$was simplified to $F = \overline{AB}$. The result was verified using a Karnaugh map and confirmed through Logisim simulation.

Experiment No-3: Simplification of $F = ABCD + A\overline{B}CD + \overline{A}BCD + AB\overline{C}D + ABC\overline{D}$

Problem Definition

Simplification of the Boolean expression $F = ABCD + A\overline{B}CD + \overline{A}BCD + AB\overline{C}D + ABC\overline{D}$ using Boolean algebra and the Karnaugh map, followed by verification of both the original and simplified circuits using logic simulation.

Objective

The main objective was to simplify a four-variable sum-of-minterms Boolean expression into its minimal Sum-of-Products (SOP) form. The simplification was verified using a Karnaugh map, and both the original and simplified circuits were implemented in Logisim. Their outputs were then tested for different input combinations to confirm that both circuits produced the same results and were functionally equivalent.

Algebraic Simplification

$$\begin{aligned}
F(A,B,C,D) &= \overline{A}BCD + A\overline{B}CD + AB\overline{C}D + ABC\overline{D} + ABCD \\
&= \overline{A}BCD + ABCD + A\overline{B}CD + ABCD + AB\overline{C}D + ABCD + ABC\overline{D} + ABCD \\
&= BCD(\overline{A} + A) + ACD(\overline{B} + B) + ABD(\overline{C} + C) + ABC(\overline{D} + D) \\
&= BCD(1) + ACD(1) + ABD(1) + ABC(1) \\
&= BCD + ACD + ABD + ABC \\
&= ABC + ABD + ACD + BCD
\end{aligned}$$

Truth Table

Minterm	A	B	C	D	F
m_0	0	0	0	0	0
m_1	0	0	0	1	0
m_2	0	0	1	0	0
m_3	0	0	1	1	0
m_4	0	1	0	0	0
m_5	0	1	0	1	0
m_6	0	1	1	0	0
m_7	0	1	1	1	1
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	0
m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	0
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	1

Table 5: Truth table for $F = ABC + ABD + ACD + BCD$

Karnaugh Map

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

Table 6: K-map for $F(A,B,C,D) = ABCD + A\overline{B}CD + \overline{A}BCD + AB\overline{C}D + ABC\overline{D}$

Grouping Analysis: The 1-cells are located at minterms $m_7, m_{11}, m_{13}, m_{14}, m_{15}$. Grouping with adjacent cells around the common minterm m_{15} ($ABCD$):

- Pair 1 (ABC): Cells m_{14}, m_{15} in row $AB = 11$, columns $CD = 10, 11$, eliminating D to give ABC.
- Pair 2 (ABD): Cells m_{13}, m_{15} in row $AB = 11$, columns $CD = 01, 11$, eliminating C to give ABD.
- Pair 3 (ACD): Cells m_{11}, m_{15} in column $CD = 11$, rows $AB = 10, 11$, eliminating B to give ACD.
- Pair 4 (BCD): Cells m_7, m_{15} in column $CD = 11$, rows $AB = 01, 11$, eliminating A to give BCD.

All four pairs are essential prime implicants. Combining them gives the minimal sum-ofproducts expression:

$$F = ABC + ABD + ACD + BCD$$

which represents the 3-out-of-4 majority voting function and confirms the algebraic simplification.

Circuit Diagram

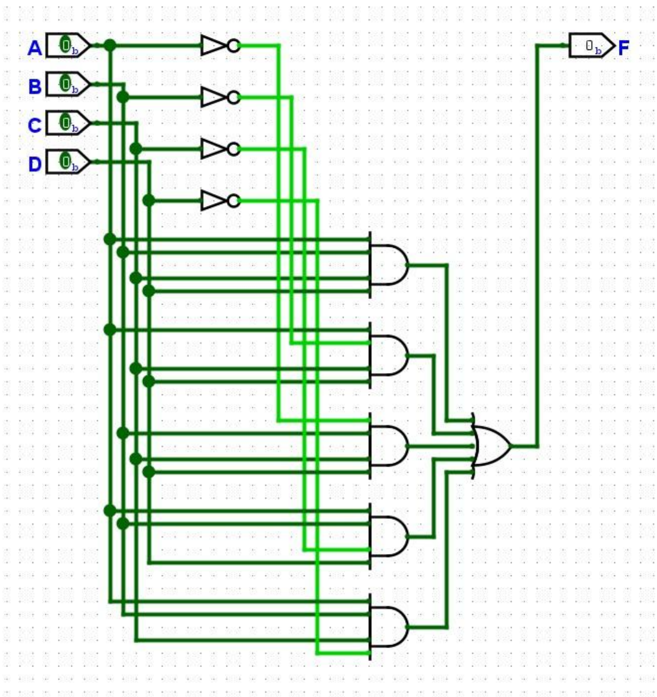


Figure 3: CKT for Lab_02_Task_03: Simplified circuit (four 3-input AND gates plus a 4-input OR gate realizing $ABC + ABD + ACD + BCD$) simulated in Logisim

Discussion

The original circuit was designed using five 4-input AND gates, resulting in a total of 25 gate inputs. Inverters were also used wherever complemented variables were required. After simplifying the Boolean expression to (F=ABC+ABD+ACD+BCD), the circuit was redesigned using four 3-input AND gates and one 4-input OR gate, reducing the total number of gate inputs to 16. Both the original and simplified circuits were implemented and tested in Logisim for all possible input combinations. The outputs matched in every case, confirming that the two circuits are functionally equivalent. Overall, the simplified circuit is more efficient because it uses fewer gates and does not require any inverters.

Conclusion

Starting from F = $ABCD + A\overline{B}CD + \overline{A}BCD + AB\overline{C}D + ABC\overline{D}$, Boolean algebra and a Karnaugh map got the expression down to F = ABC + ABD + ACD + BCD. To check that this simplification actually held up, both circuits were built in Logisim and run through every input combination — and the outputs matched every time.

Experiment No – 4: Simplification of POS Expression $F = (A + B + C + D)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + C)$

Problem Definition

Simplification of the four-variable product-of-sums (POS) Boolean expression $F = (A + B + C + D)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + C)$ using Boolean algebra and Karnaugh map techniques, followed by verification using logic simulation in Logisim.

Objective

To convert and simplify a multi-factor POS Boolean expression into its minimal two-level SOP form using algebraic identities and a 4-variable Karnaugh map, and to verify functional equivalence in Logisim.

Algebraic Simplification

Given the POS expression:

$$F = (A + B + C + D)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + C)$$

Multiplying the Second and Third Factors using Distributive Law:

$$(A + B + \overline{C})(A + \overline{B} + C) = A + (B + \overline{C})(\overline{B} + C)$$

$$= A + (B\overline{B} + BC + \overline{C}\overline{B} + \overline{C}C)$$

$$= A + BC + \overline{B}\overline{C}$$

Multiplying with the Fourth Factor $(\overline{A} + B + C)$: Using the identity $(X + Y)(\overline{X} + Z) = XZ + \overline{X}Y$:

$$(A + BC + \overline{B}\overline{C})(\overline{A} + B + C) = A(B + C) + \overline{A}(BC + \overline{B}\overline{C})$$

$$= AB + AC + \bar{A}BC + \overline{\mathrm{ABC}}$$

$Since\ AC + \bar{A}BC = AC + BC$:

$$= AB + BC + AC + \overline{\mathrm{ABC}}$$

Multiplying with the First Factor$(\boldsymbol{A} + \boldsymbol{B} + \boldsymbol{C} + \boldsymbol{D})$**:** $Writing\ (A + B + C + D) = (A + B + C) + D$

$$F = \left(AB + BC + AC + \overline{\mathrm{ABC}}\right)[(A + B + C) + D]$$

$$= \left(AB + BC + AC + \overline{\mathrm{ABC}}\right)(A + B + C) + \left(AB + BC + AC + \overline{\mathrm{ABC}}\right)D$$

$$Since\ (AB + BC + AC)(A + B + C) = AB + BC + AC\ and\ \left(\overline{\mathrm{ABC}}\right)(A + B + C) = 0$$

$$F = (AB + BC + AC) + ABD + BCD + ACD + \overline{\mathrm{ABC}}D$$

$$By\ absorption\ (AB + ABD = AB, BC + BCD = BC, AC + ACD = AC):$$

$$F = AB + BC + AC + \overline{\mathrm{ABC}}D$$

Truth Table

Minterm	A	B	C	D	F
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_2	0	0	1	0	0
m_3	0	0	1	1	0
m_4	0	1	0	0	0
m_5	0	1	0	1	0
m_6	0	1	1	0	1
m_7	0	1	1	1	1
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	1
m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	1
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	1

Truth table for $F = AB + BC + AC + \overline{\mathrm{ABC}}D$

Karnaugh Map

AB \ CD	00	01	11	10
00	0	1	0	0
01	0	0	1	1
11	1	1	1	1
10	0	0	1	1

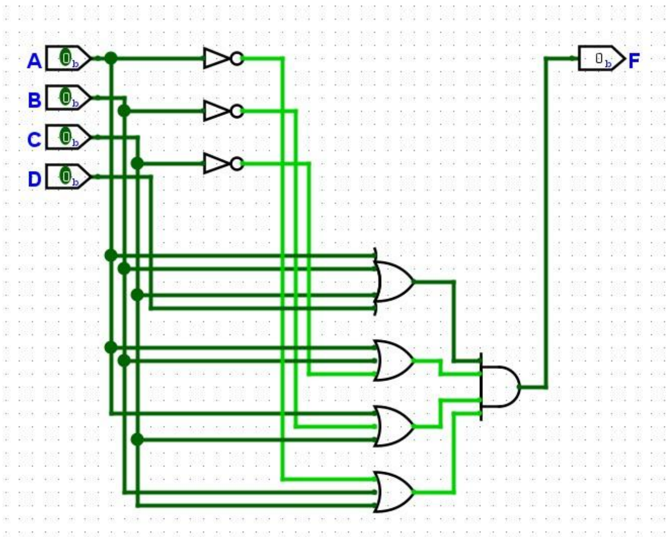
K-map for $F(A,B,C,D) = \prod M(0,2,3,4,5,8,9)$

GroupingAnalysis: The 1-cells are located at minterms $m_1, m_6, m_7, m_{10}, m_{11}, m_{12}, m_{13}, m_{14}, m_{15}$:

- Quad 1 (AB): Cells $m_{12}, m_{13}, m_{15}, m_{14}$ in row $AB = 11$, yielding AB.
- Quad 2 (BC): Cells m_6, m_7, m_{14}, m_{15} in rows $AB = 01, 11$, columns $CD = 11, 10$, yielding BC.
- Quad 3 (AC): Cells $m_{10}, m_{11}, m_{14}, m_{15}$ in rows $AB = 11, 10$, columns $CD = 11, 10$, yielding AC.

Combining these essential prime implicants results in the minimal SOP expression: $F = AB + BC + AC + \overline{ABC}D$

Circuit Diagram



Discussion

The original POS network requires one 4-input OR gate, three 3-input OR gates, three inverters, and a 4-input AND gate. Converting to minimal SOP form produces a two-level AND-OR structure requiring three 2-input AND gates, one 4-input AND gate, three inverters, and one 4-input OR gate. Both circuits were simulated in Logisim. The output evaluated to 0 for the 7 maxterm combinations ($M_0, M_2, M_3, M_4, M_5, M_8, M_9$) and to 1 for all remaining 9 combinations, confirming full functional equivalence.

Conclusion

The POS expression was simplified to its minimal SOP form $F = AB + BC + AC + \overline{ABC}D$. The result was confirmed using the Karnaugh map and validated by Logisim simulation.

Experiment No-5: Simplification of Hierarchical Expression $F = \left(A + \bar{B}(C + D)\right) \cdot \left(\bar{A} + B\left(\bar{C} + \bar{D}\right)\right)$

Problem Definition

Simplification of the hierarchical Boolean expression $F(A, B, C, D) = \left(A + \bar{B}(C + D)\right) \cdot \left(\bar{A} + B\left(\bar{C} + \bar{D}\right)\right)$ using Boolean algebra and Karnaugh map techniques, followed by verification using logic simulation in Logisim.

Objective

To simplify a multi-level factored Boolean expression into its minimal sum-of-products form using algebraic expansion and distributive laws, verify the prime implicants via a 4-variable Karnaugh map, and confirm operational behavior through Logisim simulation.

Algebraic Simplification

Given the expression:$F(A, B, C, D) = \left(A + \bar{B}(C + D)\right) \cdot \left(\bar{A} + B\left(\bar{C} + \bar{D}\right)\right)$

1. **Expanding the Expression Term-by-Term:**

$$F = A \cdot \bar{A} + A \cdot B\left(\bar{C} + \bar{D}\right) + \bar{B}(C + D) \cdot \bar{A} + \bar{B}(C + D) \cdot B\left(\bar{C} + \bar{D}\right)$$

2. **Evaluating Each Component Term:**
   - $A \cdot \bar{A} = 0 (Complement/Inverse Law)$
   - $A \cdot B\left(\bar{C} + \bar{D}\right) = AB\left(\bar{C} + \bar{D}\right) = AB\bar{C} + AB\bar{D}$
   - $\bar{B}(C + D) \cdot \bar{A} = \bar{A}\bar{B}(C + D) = \bar{A}\bar{B}C + \bar{A}\bar{B}D$

$$\bar{B}(C + D) \cdot B\left(\bar{C} + \bar{D}\right) = \left(\bar{B}B\right)(C + D)\left(\bar{C} + \bar{D}\right) = 0 \cdot (C + D)\left(\bar{C} + \bar{D}\right) = 0\left(since\,\bar{B}B = 0\right)$$

3. **Summing Non-Zero Terms:**

$$F = AB\bar{C} + AB\bar{D} + \bar{A}\bar{B}C + \bar{A}\bar{B}D$$
$$= AB\left(\bar{C} + \bar{D}\right) + \bar{A}\bar{B}(C + D)$$

Truth Table

Minterm	A	B	C	D	F
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_2	0	0	1	0	1
m_3	0	0	1	1	1
m_4	0	1	0	0	0
m_5	0	1	0	1	0
m_6	0	1	1	0	0

m_7	0	1	1	1	0
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	0
m_{11}	1	0	1	1	0
m_{12}	1	1	0	0	1
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	0

Truth table for $F = AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D$

Karnaugh Map

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	0	0
11	1	1	0	1
10	0	0	0	0

K-map for $F(A,B,C,D) = \sum m(1,2,3,12,13,14)$

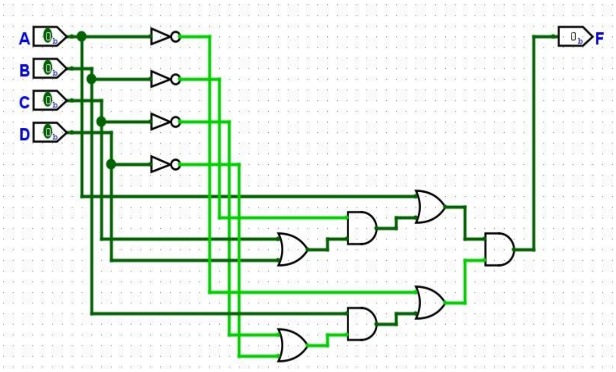
Grouping Analysis: The 1-cells are located at minterms $m_1, m_2, m_3, m_{12}, m_{13}, m_{14}$:

- Pair 1 ($\overline{A}\overline{B}D$): Cells m_1, m_3 in row $AB = 00$, columns $CD = 01, 11$, eliminating C to give $\overline{A}\overline{B}D$.
- Pair 2 ($\overline{A}\overline{B}C$): Cells m_2, m_3 in row $AB = 00$, columns $CD = 11, 10$, eliminating D to give $\overline{A}\overline{B}C$.
- Pair 3 ($AB\overline{C}$): Cells m_{12}, m_{13} in row $AB = 11$, columns $CD = 00, 01$, eliminating D to give $AB\overline{C}$.
- Pair 4 ($AB\overline{D}$): Cells m_{12}, m_{14} in row $AB = 11$, columns $CD = 00, 10$, eliminating C to give $AB\overline{D}$.

These four 2-cell groupings form the complete set of essential prime implicants. Thus, the minimized sum-of-products expression is:

$$F = AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D$$

Circuit Diagram



Minimized circuit ($AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D$) simulated in Logisim

Discussion

The original expression is a three-level hierarchical network consisting of four OR gates, three AND gates, and four inverters. Expanding the terms eliminates complementary contradictions

($\overline{A}A = 0$ and $\overline{B}B = 0$), transforming the circuit into a standard two-level AND-OR structure. The minimized SOP circuit requires four 3-input AND gates, four inverters, and one 4-input

OR gate. Simulation in Logisim verified that the output is logic 1 precisely for the six minterms ($m_1, m_2, m_3, m_{12}, m_{13}, m_{14}$), confirming functional equivalence.

Conclusion

The hierarchical Boolean expression $F = \left(A + \overline{B}(C + D)\right) \cdot \left(\overline{A} + B\left(\overline{C} + \overline{D}\right)\right)$ was successfully simplified to its minimal SOP form $AB\overline{C} + AB\overline{D} + \overline{A}\overline{B}C + \overline{A}\overline{B}D$. The result was confirmed by 4-variable Karnaugh map reduction and verified through Logisim logic simulation.